Farzaneh Fayazbakhsh

Stockholm, Sweden fa.fayazbakhsh@gmail.com 079-346 08 88 fari-labwitch

Personal Statement

Interdisciplinary biomedical researcher combining bioinformatics, genomics and cancer biology with hands-on wet-lab experience. I develop reproducible R/Python workflows for sequencing and transcriptomic data, including single-cell and spatial analyses, and connect computational results to disease mechanisms, biomarkers and translational questions through experience at Karolinska Institutet, KTH and SciLifeLab.

Bioinformatics & Computational Biology Projects

Automated Raw Sequencing Data Quality Control Pipeline

- Built an automated, parallelized Bash orchestration workflow designed for execution on High-Performance Computing (HPC) clusters via SLURM scheduling.
- Deployed FastQC and fastp dynamically inside isolated, version-controlled Singularity and Apptainer containers to automate raw sequence pairing, quality checking, and adapter trimming.
- Implemented multi-core processor allocation inside container runtimes to optimize throughput when executing string substitutions and batch file calculations.
- Orchestrated MultiQC container tasks to aggregate and output comprehensive data quality matrices, tracking processing adjustments across all sample cohorts simultaneously.

Tumor Microenvironment Profiling: scRNA-Seq & Spatial Transcriptomics

- Engineered end-to-end transcriptomic pipelines in R using Seurat and SCTransform to map high-dimensional gene expression across breast cancer tissue cohorts.
- Executed data pre-processing, Quality Control (QC), linear (PCA), and non-linear (t-SNE / UMAP) dimensionality reduction alongside graph-based clustering to resolve distinct cell populations and structural tissue features.
- Containerized both analysis environments using Docker and Docker Compose, ensuring full cross-platform compatibility and seamless runtime execution across varied infrastructure platforms.
- Managed pipeline documentation, software repositories, and strict workflow reproducibility utilizing Git and GitHub version control standards.

Computational Biology & Bioinformatics Skills

Programming & Pipeline Automation: Python, R, UNIX/Linux Shell, Nextflow (pipeline development), Git, and GitHub for version control

Genomics, High-Throughput Sequencing & Genetics: Practical experience in NGS data analysis, Whole Genome Sequencing (WGS), RNA-Seq processing, GWAS, QTL analysis, gene mapping, structural variation, metagenomics, genome annotation, and phylogenomics (MEGA, Clustal Omega)

Structural Biology, Modeling & Machine Learning: Sequence analysis, primer design, AlphaFold (protein structure prediction), PyMOL (molecular visualization), and foundational familiarity with deep learning frameworks

Infrastructure, HPC & DevOps: Hands-on experience with High-Performance Computing (HPC) clusters, containerization (Docker, Docker Compose, Apptainer/Singularity), and implementation of FAIR Research Data Principles

Research Experience

Karolinska Institute & SciLifeLab, Master's Thesis Research — 2026–present

- Investigating replication-stress-induced genomic instability at common fragile sites using targeted long-read sequencing.
- Performing neuroblastoma cell culture, drug treatments, DNA extraction, and long-range PCR.
- Developed bioinformatics workflows using Python, Linux and HPC environments for long-read sequencing analysis.
- Performed structural variant discovery, breakpoint characterization and endogenous barcode identification.

KTH Royal Institute of Technology & SciLifeLab, Research Engineer — Sweden, 2023–2024

- Independently developed Tumor-on-a-Chip models for in vitro anti-cancer drug screening.
- Optimized layer-by-layer cellulose nanofibril coatings to enhance cancer spheroid formation.

- Contributed to projects on Brain-on-a-Chip models for metabolic studies.
- Functionalized electrochemical sensors for ketone detection.

Tehran University of Medical Sciences, Master's Thesis Research — Iran, 2018–2022

- Synthesized gold nanoparticles conjugated with polyphenols.
- Developed microfluidic devices for endothelial cell culture.
- Modeled early-stage endothelial dysfunction and conducted oxidative stress assays.
- Assisted in preparing a PET/PDMS/Collagen nanofibrous membrane for microfluidic cell culture.

Gen-Iran Lab, Pathology and Cytology Intern — Tehran, Iran, 2023

- Hands-on experience in tissue preparation, fixation, embedding, and sectioning.
- Performed histological and immunohistochemical staining.

Education

M.Sc. **Swedish University of Agricultural Sciences**, Bioinformatics — 2025–present

- Highest grade (5/5) in Bioinformatics course
- Thesis: Exploring the Potential of Fragile Sites as Markers for Lineage Tracing

KTH Royal Institute of Technology, Single-Cell & Spatial Transcriptomics Data Analysis Course — 2024

M.Sc. **Tehran University of Medical Sciences**, Medical Nanotechnology — Iran, 2017–2022

- Awarded as the Top Master's Graduate (GPA: 4/4)
- Thesis: Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip

B.Sc. **Alzahra University**, Cellular and Molecular Biology (Biotechnology Specialization) — Iran, 2012–2016

Peer-Reviewed Publications

Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip — 2023
Farzaneh Fayazbakhsh, Fatemeh Hataminia, Houra Mobaleghol Eslam, Mohammad Ajoudanian, Sharmin Kharrazi, Kazem Sharifi, Hossein Ghanbari
Scientific Reports

Metabolic Assessment of iPSC-Derived Neurons Under Ketogenic and Normal Diets: Ketone Sensor Development and Comparative Analysis of Neuronal Metabolism — 2026
Rohollah Nasiri*, **Farzaneh Fayazbakhsh***, Reyhaneh Sanei, Tingting Wu, Nayere Taebnia, Rouhollah Habibey, Omid Fotouhi, John Cognetti, Volker M. Lauschke, Aman Russom, Anna Herland
CellPress

Assessing Layer-by-Layer Assembly of Cellulose Nanofibrils in Pancreatic Tumor Spheroid Formation — 2023
Negar Abbasi Aval, Ekeram Lahchaichi, Oana Tudoran, **Farzaneh Fayazbakhsh**, Rainer Heuchel, Matthias Löhr, Torbjörn Pettersson, Aman Russom
Biomedicines

Characterization and Numerical Simulation of a New Microfluidic Device for Cell Investigations — 2024
H Meslam, F Hataminia, H Asadi-Saghandi, **F Fayazbakhsh**, N Tabatabaei, Hossein Ghanbari
Frontiers in Lab on a Chip Technologies

Nanofibers in Respiratory Masks: An Alternative to Prevent Pathogen Transmission — 2022
Tabatabaei SN, Faridi-Majidi R, Boroumand S, Norouz F, Rahmani M, Rezaie F, **Fayazbakhsh F**, Faridi-Majidi R
IEEE Transactions on NanoBioscience

Wet Lab & Experimental Expertise

Cell Culture & Cancer Biology: 2D & 3D cell culture, tumor spheroid formation and organoid culture, in vitro cancer biology drug screening

Molecular Biology & Genomics Workflows: DNA extraction, downstream cellular and molecular assessments, PCR / Real-Time PCR, agarose gel electrophoresis, Agilent TapeStation analysis, flow cytometry, fluorescent microscopy

Microfluidics & Biosensors: Organ-on-a-Chip modeling, PDMS-based microfluidic systems fabrication, surface and electrochemical sensor functionalization

Histology & Pathology: Histological sample preparation, H&E staining, immunohistochemistry (IHC)

Nanotechnology & Nanomaterial Synthesis: Green synthesis of gold nanoparticles, nanofiber electrospinning, cellulose nanofibrils (CNF) coating optimization

Conference Presentations

Spheroid Formation in Cellulose Nanofibrils-Coated Microfluidic Chips — µTAS 2024, Montreal, Canada, 2024
Poster presentation (first author)

Design of a Microfluidic Electrochemical Ketone Sensor — Swedish Microfluidics in Life Science, KTH, Sweden, 2023
Oral presentation (co-author)

Investigation of Effect of Different Diets on Brain Energy Metabolism Using Organ-on-a-Chip — MPS World Summit, Berlin, Germany, 2023
Poster presentation (co-author)

Preparation of PET/PDMS/Collagen Nanofibrous Membrane Embedded in a Microfluidic Device for Organ-on-Chip Applications — The 3rd Conference on Nanofibers, Tehran University of Medical Sciences, 2022
Poster presentation (co-author)

Professional & Personal Skills

Bridging Wet & Dry Lab: Confident working in both experimental labs and computational environments, making it easy to translate biological questions into data-driven answers

Independent Problem Solver: Highly autonomous in research environments, with a proven ability to troubleshoot bottlenecks and quickly teach myself new tools to keep projects moving forward

Collaborative Team Player: Experienced in international, interdisciplinary teams such as KTH, Karolinska Institute and SciLifeLab, with a strong focus on knowledge-sharing and helping colleagues

Clear Communicator & Fast Learner: Skilled at explaining complex data clearly through peer-reviewed publications and conference presentations; highly adaptable and quick to pick up new scientific topics and skills

Teaching & Mentorship

KTH Royal Institute of Technology, Intern Supervisor — Sweden

- Guided an intern in laboratory techniques.

Volunteer Service, Science Teacher

- Taught science to underprivileged children.

Awards & Honors

Top Master's Graduate — Tehran University of Medical Sciences, 2023

LUSH Prize Candidate 2022

Shortlisted for the Young Researcher Award.

Darolfonoon Student Award 2020

For voluntary assistance in nanofiber mask production during the COVID-19 pandemic.

Top Article Award First National and Student Conference on Applied Biology, Alzahra University, Iran 2014

Prize winner.

Top 5% in Iran's Konkour for Master's Degree University Entrance Iran 2016

Ranked 37th in Iran's national Konkour for the Master's Degree University Entrance Exam.

References

Dr. Kasper Karlsson #linebreak() Department of Oncology-Pathology, Karolinska Institute #linebreak() kasper.karlsson@ki.se

Dr. Erik Bongcam Rudloff #linebreak() Department of Animal Biosciences, Swedish University of Agricultural Sciences #linebreak() erik.bongcam@slu.se

Dr. Aman Russom #linebreak() Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology #linebreak() aman.russom@scilifelab.se

Cheng-de (Ernest) Liu #linebreak() Department of Oncology-Pathology, Karolinska Institute #linebreak() chengde.liu@ki.se

Dr. Anna Herland #linebreak() Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology #linebreak() aherland@kth.se

Dr. Rohollah Nasiri #linebreak() Radiation Oncology Department, Stanford University #linebreak() rhnasiri@stanford.edu